Separate a change from its cause

Notice what changed, then ask what evidence would identify the cause.

Suggested teaching time: 30 min | Preparation: about 5 min

Introductory | Secondary and adult learners | Individual writing, pairs, and class discussion

Prerequisites: None

Students will be able to

- Distinguish an observed change from a claim about its cause.
- Identify a simultaneous change that could explain the result.
- Propose a comparison that separates two possible causes.

Prepare and gather

- One student handout per learner, or display the activities and use paper.
- Pens or pencils; no student devices required.
- Instructor guide with model answers, kept separate from the handout.
- Read the three scenarios and their model answers.
- Print the three-page handout or display one activity at a time.
- Explain that all scenarios and figures are invented for reasoning practice.

Teaching sequence

5 min	**Spot the competing changes.** Learners complete the fruit-sales warm-up. Compare the observed outcome with the manager's explanation.
10 min	**Separate lighting from patrols.** Pairs complete the street-lighting activity. Check that their proposed comparison changes lighting while keeping patrol conditions comparable.
8 min	**Test the comparison.** Discuss the supplied prompts and model answer. Ask whether another change is still tied only to the treatment group.
7 min	**Transfer the method.** Learners complete the attendance exit ticket independently. Assess the observation, alternative cause, and proposed comparison.

Timings are suggested starting points, not measured classroom completion times.

Warm-up / Why did fruit sales rise?

Student task and instructor answer

Explain these words as needed

Cause: Something that helps produce an outcome.

Comparison group: A group used to judge what might happen without the change being tested.

Random assignment: Using chance to decide which groups receive a change, helping avoid systematic differences.

Scenario

Invented scenario: A cafeteria sells more fruit in the week it displays healthy-eating posters. That week it also cuts fruit prices. The manager says the posters caused the rise.

Student output: Write an observation, another possible cause, and one sentence explaining why the posters have not been isolated.

1. State what happened without naming its cause.

2. Name the other change and explain how it could affect sales.

3. Explain why these facts cannot establish the effect of posters alone.

Student success check: Have you kept an observed rise separate from an explanation for it?

Model answer

Observation: fruit sales rose. Lower prices could encourage purchases. Because posters and prices changed together, the rise does not show how much, if any, came from the posters.

Why this meets the task: The answer identifies a relevant competing change without claiming to know which cause mattered.

Main activity / Lighting, patrols, and theft reports

Student task and instructor answer

Scenario

Invented scenario: Reported thefts on a street fall from 50 in September to 35 in October. Brighter lights are installed at the start of October, and extra evening patrols begin that same day. The mayor says the lights caused the drop.

Student output: Write the observation, a competing explanation, and a proposed comparison in three labelled parts.

1. State the observed change separately from the mayor's causal explanation.
2. Name the other change that could explain some or all of the drop.
3. Propose a comparison that would help separate the effects: describe which streets to compare and which other condition should be kept similar. Do not claim your proposed comparison has already been done.

Student success check: Does the comparison vary lighting without also giving only the newly lit streets extra patrols?

Model answer

The observation is that reports fell from 50 to 35 after two changes. Extra patrols are a rival explanation. Compare similar streets over the same period, some receiving brighter lights and others retaining existing lights, while keeping patrol levels comparable; random assignment of lighting would make the test stronger where feasible.

Why this meets the task: It identifies the simultaneous change and proposes evidence that could distinguish explanations instead of treating 'after' as 'because of'.

Exit ticket / Why did absences fall?

Student task and instructor answer

Scenario

Invented scenario: Weekly absences in a school fall from 12 to 5 after text reminders and a new bus service begin on the same day. The principal credits the reminders alone.

Student output: Write the observed change, another possible cause, and a comparison that could test reminders.

1. State the observed change without attributing a cause.

2. Name the other change and explain its possible effect.

3. Describe groups to compare and which other condition must stay comparable.

Student success check: Would your comparison give similar bus access to students with and without reminders?

Model answer

Absences fell from 12 to 5. The bus service might make attendance easier. Compare similar student groups over the same period, with comparable bus access, giving reminders to one group and not the other; random assignment where feasible would strengthen the comparison.

Why this meets the task: The answer separates the outcome from its explanation and proposes varying reminders without tying them to better bus access.

Check understanding

Mark each criterion met or needs another attempt. Accept equivalent wording; assess the reasoning, not a memorized phrase.

- The observed outcome is stated separately from its proposed cause.

- A supplied competing change is linked plausibly to the outcome.

- The proposed comparison varies the target factor while keeping the competing condition comparable.

Discussion and adaptations

Use the supplied guidance to keep the lesson focused on its reasoning skill.

Discussion prompts

Does another possible cause prove the lights had no effect?

Listen for: No. It prevents assigning the drop to lights alone. Lighting, patrols, both, or other factors might matter.

Why compare streets during the same period?

Listen for: Shared timing helps reduce differences such as seasonal changes. It does not guarantee that the streets are otherwise comparable.

Common confusions

After a change means because of that change. Timing can support a causal account, but simultaneous changes and missing comparisons must be considered.

Finding a rival explanation disproves the first one. A plausible rival shows uncertainty; further evidence is needed to separate their contributions.

Any two groups make a fair test. Groups should be comparable in relevant ways. Random assignment helps but does not excuse inconsistent measurement.

Adapt the lesson

More support: Use three headings: what changed, what else changed, what to compare. Let learners say their comparison aloud before writing it. Expected example: fruit sales rose; prices also fell; compare poster and no-poster settings with comparable prices.

More challenge: Ask: If brighter-light streets also get more patrols in the proposed comparison, what remains unresolved? Expected response: the effects remain mixed, so a difference cannot be attributed to lighting alone.

Next practice

https://xixteen.com/learners/?practice=causation#workbench